Fairness Audit – Income Prediction Model

8 November 2026 · Tommaso Perilli

1. Summary

Verdict: Not approved

The model predicts whether a person earns more than $50,000 a year, in a loan-offer scenario: a high-risk use under the AI Act. Accuracy is 85.5%, but overall recall (59.8%) is just below the 0.60 threshold: the model misses about 4 in 10 high-income people.

The most serious problem is fairness. At equal high income, the model recognises women less often than men (recall 54.2% vs 60.9%) and Black people less often than White people (53.7% vs 60.3%), even though it does not know sex or ethnicity. No fairness criterion is met.

Release is not approved until the high-priority actions in section 7 are completed and a new audit is run against the same criteria.

2. System and context of use

Item	Description
What the system does	Predicts whether a person earns more than $50,000 a year
Intended use	A bank using the model to decide which customers receive a loan offer
Who is affected by the decision	People applying for a loan, or potential customers of the bank
AI Act classification	High risk: assessment of the creditworthiness of natural persons
Model type	Logistic regression (a simple, widely used model)

A model error has concrete consequences for people. Someone with a high income who is wrongly rejected does not receive the loan offer. If this happens more often to one group, for example women, the error becomes systematic discrimination. This is why the AI Act classifies credit assessment as a high-risk use and requires fairness to be checked before release.

3. Data (AI Act, art. 10)

Source: public Adult Income dataset, 1994 US Census. Total people: 48,842, of whom 23.9% have a high income (over $50,000 a year).

Consistency check: the totals by sex and by ethnic group both add up to 48,842.

Sex	People	Share of total	People in test set (estimate, 30%)	Share with high income
Men	32,650	66.8%	about 9,800	30.4%
Women	16,192	33.2%	about 4,860	10.9%

Ethnic group	People	Share of total	People in test set (estimate, 30%)	Share with high income
White	41,762	85.5%	about 12,530	25.4%
Black	4,685	9.6%	about 1,410	12.1%
Asian-Pac-Islander	1,519	3.1%	about 460	26.9%
Amer-Indian-Eskimo	470	1.0%	about 140	11.7%
Other	406	0.8%	about 120	12.3%

Observations

- **Under-represented groups:** Asian-Pac-Islander (3.1%), Amer-Indian-Eskimo (1.0%) and Other (0.8%). In the test set they will have about 460, 140 and 120 people: measurements for Amer-Indian-Eskimo and Other will be unreliable. Women are also under-represented: 33% of the dataset, against roughly half of the population.
- **Differences already in the data:** the share with high income is 30.4% among men and 10.9% among women (2.8 times). By ethnic group it is 25.4% (White) and 26.9% (Asian-Pac-Islander), against 12.1% (Black), 11.7% (Amer-Indian-Eskimo) and 12.3% (Other): about double.
- **Fact and risk:** these differences are in the data, not yet in the model. The risk is that the model learns and amplifies them; steps 5 and 6 test this.
- **Reference for section 5:** with 23.9% high incomes, a model that always answered "no" would reach about 76% accuracy.
- **Possible proxies ("hidden clues"):** even without Sex and Race, the relationship column reveals sex, because it contains the values Husband and Wife. Other columns may reflect sex or ethnicity: marital-status, occupation and hours-per-week for sex, native-country for ethnicity.
- **Data quality:** workclass, occupation and native-country contain missing values; the information is self-reported in the census.
- **Age and context of the data:** the data are from 1994 and US-only. $50,000 then is worth more than twice as much today, the labour market has changed, and the ethnic categories are those of the US census. For a European bank they would not represent the context of use, as AI Act art. 10 requires.

4. Test method and acceptance criteria (AI Act, arts. 9 and 15)

How it was tested

- Data split: 70% training, 30% test, never seen by the model during training.
- Sex and ethnicity **excluded** from the model inputs, but used to measure results by group.
- Metrics: accuracy, precision, recall, selection rate.
- Groups analysed: sex and ethnic group.

Acceptance criteria

The criteria were defined **before** looking at the results by group (steps 5 and 6).

Criterion	Threshold	Rationale
Minimum overall accuracy	at least 0.80	Must clearly beat the 76% of the "lazy" model that always answers "no" (see section 3). Otherwise the model adds no value.
Minimum overall recall	at least 0.60	The model must recognise at least 6 in 10 high-income people. Below this, it would exclude too many people entitled to a loan offer.
Maximum recall gap between groups	no more than 0.05	The most important criterion: at equal real income, every group should be recognised with similar probability. A larger gap means treating a group worse under equal conditions.
Minimum selection rate ratio between groups	at least 0.80	The "four-fifths rule". Used as a warning sign, not a verdict: the share of high incomes already differs between groups in the data (section 3), so part of the gap is expected. Read together with recall.
Minimum group size for a reliable measurement	at least 100 people in the test set, of whom at least 30 with high income	Recall is computed only on high-income people. Amer-Indian-Eskimo and Other will have about 15 each in the test set (140 × 11.7% and 120 × 12.3%): too few for a stable measurement.

Decision rule

- All criteria met: release approved.
- Overall criteria met, but a group criterion not met: approved only with conditions (corrective actions and monitoring, section 7).
- An overall criterion not met: release not approved.
- A group below the reliability threshold is marked as **not assessable**, not as passed, and is declared among the limitations (section 6).

5. Results

5.1 Overall results (step 4)

Metric	Value	What it means, in plain words
Accuracy	85.5%	Out of 100 people in the test set, the model classifies about 86 correctly. This is above the 76% of the "lazy" model that always answers "no": the model adds value.
Precision	74.6%	When the model says "high income", it is right 3 times out of 4. One loan offer in 4 would go to someone who actually has a low income.
Recall	59.8%	Out of 100 high-income people, the model recognises about 60. The other 40 would not receive the loan offer.
False negatives	1,408	High-income people the model rejects: they would not receive the offer despite being entitled to it. This is the error that harms the person.
False positives	715	Low-income people the model treats as high-income: they would receive an offer they may not be able to sustain. This is the error that mainly costs the bank.

The 85.5% accuracy must be read with caution: only 23.9% of people have a high income, so a model that always answered "no" would already reach about 76%. The real gain is about 10 points. The real limitation shows in recall: 59.8%, just below the 0.60 threshold set in section 4. The model misses about 4 in 10 high-income people (1,408 in the test set).

5.2 Results by sex (step 5)

Sex	People in test set	Accuracy	Precision	Recall	Selection rate
Men	9,796	81.8%	74.2%	60.9%	24.7%
Women	4,857	92.9%	76.9%	54.2%	8%
Maximum difference	–	11.1%	2.7%	6.6%	16.7% (ratio 0.32)

Women's recall is 54.2% against 60.9% for men: at equal high income, the model recognises 54 women out of 100 and 61 men out of 100. The gap (6.6 points) exceeds the 5-point threshold. The selection rate is 8% against 24.7%, a ratio of 0.32 against the required 0.80; in the data the ratio was 0.36, so the model amplifies the gap. Women's higher accuracy (92.9%) is not an advantage: with 89% low incomes, answering "no" almost every time is enough. Precision is similar (76.9% vs 74.2%). Both groups are large enough for reliable measurements. Removing the Sex column was not enough: the model treats women worse at equal income, and the gap exceeds the criteria set.

5.3 Results by ethnic group (step 6)

Ethnic group	People in test set	Accuracy	Precision	Recall	Selection rate
White	12,455	84.9%	75.1%	60.3%	20.3%
Black	1,471	91.2%	70.1%	53.7%	9.8%
Asian-Pac-Islander	461	82.0%	69.3%	62.2%	24.7%
Amer-Indian-Eskimo (not assessable)	138	90.6%	83.3%	47.6%	8.7%
Other (not assessable)	128	89.8%	55.6%	35.7%	7%
Maximum difference	–	9.2%	27.8%	26.5%	17.7%
Maximum difference, reliable groups	–	9.2%	5.8%	8.5%	14.9% (ratio 0.40)

Amer-Indian-Eskimo and Other cannot be assessed: in the test set they have only 21 and 14 high-income people, below the threshold of 30. For Other, one more person recognised would move recall by about 7 points. For this reason the overall maximum difference (26.5 recall points) is not solid evidence, and the comparison is based on the three reliable groups: White, Black and Asian-Pac-Islander.

Among these, recall ranges from 62.2% (Asian-Pac-Islander) and 60.3% (White) down to 53.7% (Black): an 8.5-point gap, above the 5-point threshold. At equal high income, Black people are recognised less often. The selection rate is 9.8% for the Black group against 20.3% for the White group (ratio 0.48) and 24.7% for Asian-Pac-Islander (ratio 0.40), below the required 0.80. Compared with the White group, the model reflects the gap already present in the data (ratio 0.48 there too); compared with Asian-Pac-Islander it amplifies it slightly (0.40 vs 0.45 in the data).

The Black group's higher accuracy (91.2%) is not an advantage: it comes from the low share of high incomes, as for women. Precision is similar across groups (between 69.3% and 75.1%). Here too, removing the Race column was not enough: the recall gap criterion is not met.

5.4 Comparison with the criteria

Criterion	Threshold	Result	Met?
Minimum overall accuracy	at least 0.80	85.5%	Yes
Minimum overall recall	at least 0.60	59.8%	**No** (by 0.2 points)
Maximum recall gap – sex	no more than 5 points	6.6 points (men 60.9%, women 54.2%)	**No**
Maximum recall gap – ethnic group (reliable groups)	no more than 5 points	8.5 points (Asian-Pac-Islander 62.2%, Black 53.7%)	**No**
Minimum selection rate ratio – sex	at least 0.80	0.32 (0.36 in the data)	**No**
Minimum selection rate ratio – ethnic group (reliable groups)	at least 0.80	0.40 (0.45 in the data)	**No**
Group reliability	at least 100 people in the test set, of whom 30 with high income	Assessable: men, women, White, Black, Asian-Pac-Islander. Not assessable: Amer-Indian-Eskimo (21 with high income) and Other (14)	Partial

Outcome under the decision rule (section 4): an overall criterion is not met (recall), and no group criterion is met. Release is therefore **not approved**.

Was removing sex and ethnicity enough? No. Even without the Sex and Race columns, the model recognises women less often (6.6 recall points lower than men) and Black people less often (6.6 points lower than the White group). For sex it even amplifies the gap already present in the data (selection rate ratio 0.32 vs 0.36). The information comes back through other columns, such as relationship, which contains the values Husband and Wife.

6. Limitations of the analysis

The analysis shows differences in treatment between groups, but it has the following limitations, to keep in mind when reading the results.

- **Groups with few cases.** Amer-Indian-Eskimo (138 people in the test set, 21 with high income) and Other (128 people, 14 with high income) cannot be assessed: one person more or less moves their recall by several points. Black and Asian-Pac-Islander also have limited numbers (about 190 and 130 high-income people), so their gaps carry an uncertainty of a few points. The gap between men and women, measured on larger groups, is the most solid.

- **A single data split.** The test used a single training/test split. With a different split the numbers could change slightly; the test was not repeated.
- **1994 data, US only.** The $50,000 threshold is not adjusted for inflation, the labour market has changed, and the ethnic categories are those of the US census. The results do not transfer to a European bank today.
- **The label is an approximation.** The dataset measures income, not creditworthiness: the loan scenario is hypothetical, and "high income" is only a simplified indicator.
- **One model, one threshold.** Only a logistic regression was tested, with the default decision threshold. A different model, or a different threshold, could give different results.
- **Only sex and ethnicity, analysed separately.** Other groups (age, country of origin) were not analysed, nor were combinations, for example Black women, who could face a combined effect.
- **The proxy is a hypothesis.** The role of the relationship column is plausible but not verified: the column was not removed to re-measure the model.
- **Missing values not analysed.** It was not checked whether missing values in workclass, occupation and native-country affect some groups more than others.

7. Recommendations and verdict (AI Act, art. 72)

Before release

Recommended action	Problem it addresses	Priority
Collect more data on under-represented groups (women, Black, Asian-Pac-Islander, Amer-Indian-Eskimo, Other) and retrain	Lower recall for women and Black people; two groups not assessable	High
Check the role of relationship and the other proxy columns: remove or transform them, then re-measure	Bias persists even without Sex and Race	High
Try bias mitigation techniques (for example Fairlearn's fairness constraints) and re-measure against the same criteria	Recall gaps above the 5-point threshold	High
Use recent data from the real context of use (European customers today)	1994 US-only data (AI Act, art. 10)	High, before any real use
Add human review of rejected people close to the decision threshold	1,408 false negatives in the test set (AI Act, art. 14)	Medium
Repeat the test on several data splits	A single training/test split	Medium
Analyse combinations of groups (for example Black women) and other groups such as age	Analysis limited to sex and ethnicity, separately	Medium

After release: what to monitor

What to measure	How often	Alert threshold
Overall recall	Monthly	Below 0.60
Recall by sex and by ethnic group	Monthly	Gap above 5 points
Selection rate ratio between groups	Monthly	Below 0.80, or worse than the ratio already in the data
Composition of incoming data (shares by group, missing values)	Quarterly	Significant shift from the training data (drift)
Full retest against the section 4 criteria	At every retraining	Any criterion not met blocks the release

Final verdict

Not approved. Overall recall (59.8%) is below the 0.60 threshold, and no fairness criterion is met: at equal high income, the model recognises women less often (6.6 points lower than men) and Black people less often (6.6 points lower than the White group). The model can be re-evaluated after the high-priority actions, with a new audit against the same criteria.